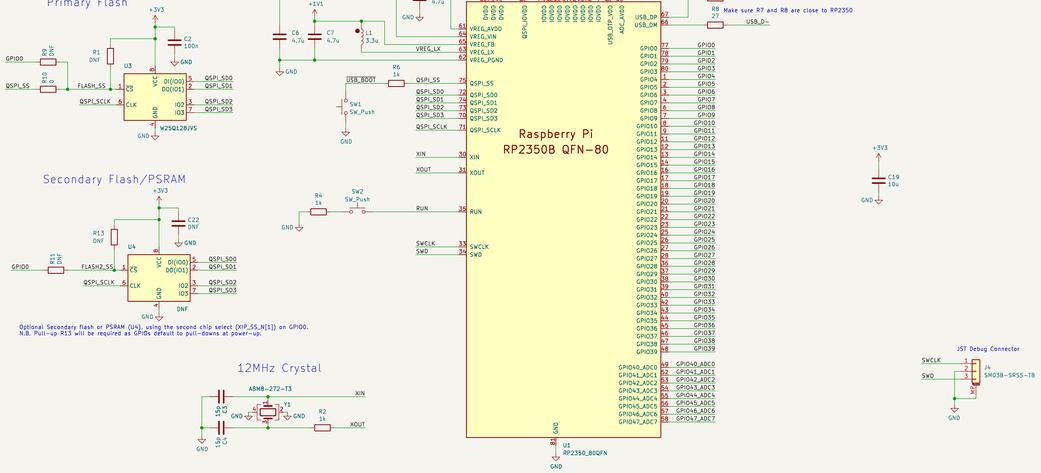
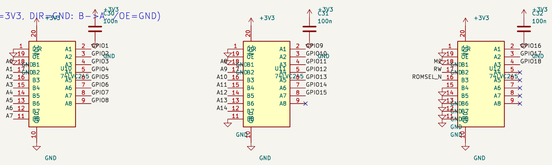
=== Famicom bus interface (FC-EXPSYNTH additions) ===

J10: console-side gold fingers (1-30 front / 31-60 back)

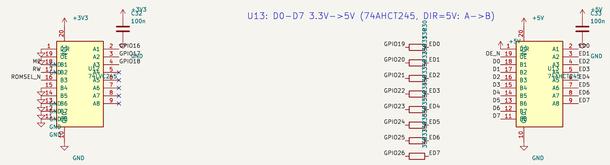
J11: 60-pin slot for the expansion cartridge (pin n -> pin n: /ROMSEL gated)



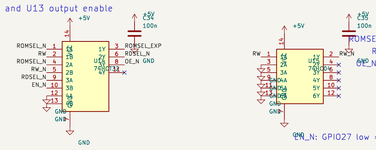
U10-U12: 5V bus -> 3.3V (74LVC245, VCC=3V3, DIR=GND: B->A; /OE=GND)



U13: D0-D7 3.3V->5V (74AHCT245, DIR=5V: A->B)



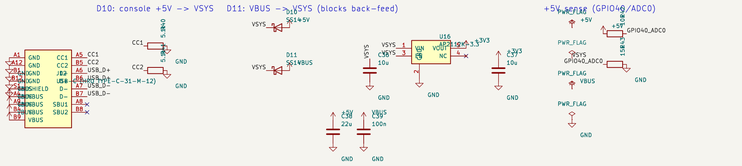
U14/U15: /ROMSEL gating and U13 output enable



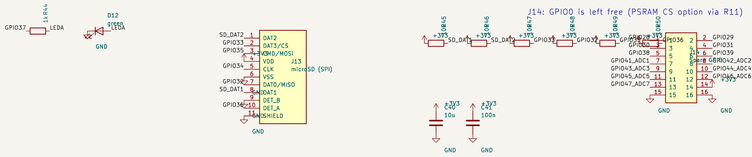
ROMSEL_EXP = ROMSEL_N OR R/W (J11 pin44)
RDSEL_N = ROMSEL_N OR NOT R/W
EN_N = RDSEL_N OR EN_N (U13 /OE)

EN_N: GPIO27 low = U13 enabled. 10k pull-up keeps it disabled at reset.

Power: console +5V and USB VBUS -> Schottky OR -> VSYS -> 3.3V LDO (IOVDD is up whenever console 5V is present)



Status LED, microSD (SPI0: GPIO32-35, CD=GPIO36), spare header



Takara (RP2350B core: Raspberry Pi Ltd minimal design REV2)	
Sheet: /	
File: FC-EXPSYNTH.kicad_sch	
Title: FC-EXPSYNTH: Famicom expansion-audio relay cartridge (RP2350B)	
Size: A1	Date: 2026-11-03
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4	Rev: E0
	Id: 1/1